

Warm-up

- (1) Write $\frac{1}{3} + \frac{1}{2} + \frac{3}{5} + \frac{2}{3} + \frac{5}{7} + \frac{3}{4} + \frac{7}{9} + \frac{4}{5} + \frac{9}{11} + \cdots$ in summation notation.
- (2) Does the series from (1) converge or diverge?
- (3) Compute $\sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n$.

Medium

Compute the following series:

$$(1) \sum_{n=1}^{\infty} \frac{2^{n+1}}{3^{2n}} \quad (2) \sum_{n=1}^{\infty} \ln \left(\frac{n+4}{n+3} \right) \quad (3) \sum_{n=1}^{\infty} \frac{2^{2n+1}}{3^n} \quad (4) \sum_{n=1}^{\infty} \frac{2}{n^2 + 5n + 6}.$$

- (5) Suppose the k th partial sum of $\sum_{n=1}^{\infty} a_n$ is $s_k = \ln \left(\frac{6k+10}{k} \right)$. What is a_1 ? What about a_n for $n > 1$? Compute $\sum_{n=1}^{\infty} a_n$.

Stretch

- (1) Compare areas to verify that $\frac{1}{k+1} < \ln(k+1) - \ln(k) < \frac{1}{k}$ and $\ln(k+1) < \sum_{n=1}^k \frac{1}{n}$.
- (2) Use the above and the monotone convergence theorem to demonstrate that

$$a_k = \left(\sum_{n=1}^k \frac{1}{n} \right) - \ln(k+1)$$

approaches some number. (This number is called the Euler-Mascheroni constant, and it is not currently known whether it is rational or irrational.)